

IntelliCon '26 · Gate 1 Submission: Problem & Proof

Project: ResQGrid AI -- Intelligent Emergency Resource Network | <https://web-two-mu-edisuy5tdl.vercel.app>

Event: IntelliCon '26 Buildathon	Gate: 1 -- Problem & Proof (25% of score)	Domain: Public Services / Disaster Logistics	Due: 26 September 2026, 11:59 PM
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One-line hook: During monsoon disasters, Sri Lanka's emergency duty officers coordinate rescues via paper logs, overloaded phone hotlines, and WhatsApp voice notes -- with no automated triage, no live resource visibility, and no explainable priority -- while trapped victims wait hours for help that may never arrive.

1. The Problem & Its Domain

Domain: **Public Services / Disaster Response Logistics** | Word count: **138 words** (limit: 150)

During monsoon flood seasons, Sri Lanka's Emergency Operations Centres (EOCs) at District Disaster Management Coordinating Units (DDMCUs) face cascading information chaos. Rescue requests arrive simultaneously as panicked phone calls on the 117 hotline, WhatsApp voice notes, and radio chatter -- all unstructured, duplicated, and unverified. Duty officers must manually decode what is happening, where victims are located, how serious their injuries are, and which boats or ambulances are currently free. Under extreme cognitive overload, critical decisions are made on gut feel: rescue units dispatched down flooded impassable roads, elderly patients queued behind non-urgent callers, and high-priority pockets left untouched because no one mapped them. The November 2025 Sri Lanka floods (330+ deaths) and the 2016 floods (300,000+ displaced) share the same root bottleneck: not lack of resources, but inability to coordinate them fast enough.

2. Who Exactly Has This Problem

Not 'everyone in Sri Lanka' -- two specific named operational roles who bear legal accountability for life-or-death decisions:

- * **Primary -- EOC Duty Operations Officer** at District Disaster Management Coordinating Units (DMC Sri Lanka) in the five highest-risk flood districts: Colombo, Gampaha, Kalutara, Ratnapura, and Galle. Works 12-hour disaster shifts managing 4-6 simultaneous phone lines, volunteer WhatsApp groups, and walkie-talkies -- manually logging reports on paper slips with zero spatial deduplication and zero automated priority ranking.
- * **Secondary -- Tactical Dispatch Officers** at Sri Lanka Navy RABS boat squadrons, 1990 Suwa Seriya ambulance coordination desks, and Sri Lanka Red Cross Society (SLRCS) branch emergency teams -- who require confirmed GPS coordinates, patient medical urgency indicators, and safe passable route intelligence before deploying any rescue asset.

3. Proof the Problem Is Real

Route A (scores higher): **5 stakeholder conversations + 32-person survey** + Route B: **verifiable empirical records** (238 words, limit 300)

Route A -- Primary Field Conversations & Survey

Person / Role	Key Finding	Direct Quote / Observation
Former DDO, Gampaha DS	Duplicate call overload during Kelani River flooding	"8 people call about the same rooftop while an isolated family 500 m away gets no boats."
Navy RABS Coxswain, Kelani Basin	Blind dispatch into submerged obstacles & live power cables	"We were sent to Sedawatta by dinghy -- found live snapped cables blocking the canal entrance."
1990 Suwa Seriya EMT, Colombo	No medical urgency data in dispatch slips	"We don't know if the victim is hypothermic or just wet until the ambulance reaches the cut-off point."
Volunteer Lead, Kolonnawa	WhatsApp forward loops -- duplicate food drops	Multiple boats dropped food to the same accessible temple; Biyagama interior cut off 48 hours.
SLRCS Branch Coordinator, Kalutara	No audit trail for dispatch decisions	"After every flood, donors ask why we sent that boat there. We have no immutable record."

Survey -- 32 respondents (volunteers, EMTs, govt staff, survivors)	93.8% cite info-chaos as #1 bottleneck 87.5% have no live resource visibility Avg manual triage: 4.8 min/call 81.3% dispatched into impassable routes 96.9% demand human approval before any AI dispatch	Survey administered 22-25 Sep 2026. Respondents: disaster volunteers, medical first responders, govt administrative staff, and flood survivors from Colombo & Gampaha districts.
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Route B -- Verifiable Empirical Evidence (238 words)

Sri Lanka's World Bank Disaster Risk Profile confirms annual flood damage for over 40 consecutive years, with annualised losses exceeding \$313 million. The November 2025 floods and mudslides killed 330+ people across central and western provinces (BBC, DMC Situation Reports). The May 2016 Kelani River floods displaced 300,000+ (UN OCHA / IOM). Post-disaster parliamentary reviews in both events confirm that rescue delays -- not absence of personnel -- caused peak mortality. Three verified structural failures recur each season: **(1) Intake Asynchrony** -- the 117 hotline receives up to 12,000 calls/day at flood peak, dropping over 70% unanswered; citizens shift to WhatsApp where messages lack GPS coordinates or medical urgency flags. **(2) Priority Blindness** -- incoming calls are processed First-Come-First-Served or via political pressure; an elderly diabetic trapped at chest-height water waits behind a non-urgent caller. **(3) Resource Friction** -- dispatchers must make 5-8 individual phone calls to verify whether a Navy dinghy or 1990 ambulance is free, fuelled, and within passable range -- with zero visibility into flooded road closures. In December 2024, the Sri Lanka Red Cross Society documented that duplicate food drops in accessible areas and ignored interior pockets resulted directly from absence of a centralised real-time coordination tool. The bottleneck is verifiable, quantified, and annually recurring.

4. Market Size: TAM, SAM, SOM

Assumptions and sources written down explicitly per Buildathon Guide requirements.

Tier	Figure	Scope	Reasoning & Source (We estimated X based on [source], assuming Y)
TAM	\$135.8 B	Global Disaster & Incident Management Market (2026)	We estimated \$135.8B based on the MarketsandMarkets Incident & Emergency Management Report (2024), assuming 6.8% CAGR from \$107B in 2023, covering global public-safety CAD, emergency telematics, NGO crisis coordination, and civil protection software.
SAM	\$420 M / yr	South & Southeast Asia climate-vulnerable EOC software market	We estimated \$420M/yr based on ADB Regional Disaster Risk Financing data and UN ESCAP Asia-Pacific Disaster Report, assuming ~14,000 national, provincial, and district disaster coordination centres across Sri Lanka, Bangladesh, India, Nepal, Philippines, Vietnam, and Indonesia adopting coordination software at ~\$30,000/yr.
SOM	\$1.85 M (Y1: \$180 k ARR)	Sri Lanka national EOC modernisation; Year-1 beachhead: Western Province	We estimated \$1.85M based on Sri Lanka National Disaster Management Plan and Ministry of Defence EOC budget allocations, assuming 25 District Secretariat EOCs + 9 Provincial EOCs + Tri-Forces / 1990 / Red Cross wings at \$45,000/yr per district cluster. Year-1 target: \$180k ARR from Colombo, Gampaha, and Ratnapura.

5. Who Else Is Solving This

At least three alternatives, including 'nothing -- they use WhatsApp and a notebook':

Alternative	How it works	Critical gaps	Why ResQGrid AI is different
(1) Nothing -- WhatsApp groups & paper notebooks (current default for Sri Lanka EOCs)	Duty officers write reports on paper slips, coordinate via personal WhatsApp and phone calls.	Over 70% call-drop rate; zero deduplication; no live map; no priority scoring; messages buried in group chat feeds.	Structured intake in ~2 seconds; spatial deduplication; live operational map; transparent 0-100 priority score with reason codes.
(2) Legacy government portals (DEWN early-warning, Sahana Eden, DMC static registries)	DEWN broadcasts SMS warnings. Sahana Eden provides post-disaster camp and supply management forms.	Built for post-disaster census or broadcast alerts -- not live tactical dispatch. No AI triage, no real-time routing, no responder lifecycle tracking.	Real-time dispatch engine; dynamic hazard-route avoidance; mobile responder tracking on PWA / Android APK.

(3) Enterprise CAD platforms (Hexagon OnCall, Motorola PremierOne, Everbridge)	High-end proprietary 911 dispatch suites used by North American and European municipal emergency services.	\$500k-\$2M upfront; proprietary hardware; closed-source; no Sinhala/Tamil NLP; incompatible with Sri Lanka's distributed EOC structure.	Open-source core; runs on any smartphone; affordable cloud SaaS; explainable AI; full offline PWA mode.
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6. The Solution

Word count: 224 words (limit: 250)

ResQGrid AI is an intelligent emergency resource coordination network that turns disaster information chaos into structured, actionable operational decisions. The platform provides a unified web-based command dashboard for dispatchers and a lightweight Progressive Web App / native Android interface for field responders and citizens.

When an incident is reported in plain unstructured language, a dual-engine triage ensemble -- a local deterministic NLP lexicon combined with an optional large language model -- extracts a typed emergency schema (incident type, severity, casualty count, medical need, vulnerable-person flags) in approximately two seconds. The lexicon always runs; the LLM is an optional second opinion.

An explainable, deterministic priority engine scores each incident across six weighted factors: Life Risk (30%), Medical Urgency (20%), People at Risk (15%), Environmental Risk (15%), Time Sensitivity (10%), and Evidence Confidence (10%). Every score decomposes into labelled reason codes -- no black box.

A global resource-matching engine uses the Kuhn-Munkres (Hungarian) algorithm to find the optimal incident-to-resource assignment, penalising routes through reported hazard zones.

The core operating principle is: **AI recommends; humans approve**. No resource is dispatched autonomously. Dispatchers review, approve, or reject every recommendation; every AI output and human decision is written to an immutable audit log. When a hazard is reported mid-operation -- a road floods, a downed power line appears -- the system re-plans and re-proposes; the human re-approves.

7. Why Does This Need AI?

Word count: 87 words (limit: 100) -- honest, no oversell

AI solves exactly one bottleneck that cannot be fixed with rules alone: extracting a structured emergency schema from panicked, grammatically chaotic free-text in under two seconds -- roughly 10x faster than a trained call-handler. A lexicon alone misses synonyms and context; the LLM covers the long tail. Everything else -- priority scoring, resource matching, route penalty, audit logging -- is deterministic and fully explainable. AI is advisory; it never writes to the dispatch log without a human signature.

8. Tech Stack & Three-Week Plan

Technical Stack

Layer	Technologies
Frontend	Next.js 14 (App Router) · TypeScript · Tailwind CSS · Leaflet / MapLibre GL (live maps) · Capacitor (Android APK) · Service Worker PWA + IndexedDB offline outbox
Backend API	FastAPI (Python 3.11) · Pydantic v2 schema validation · SQLAlchemy 2.0 ORM · Alembic migrations · JWT auth + 4-role RBAC (Citizen / Responder / Dispatcher / Admin)
Database	PostgreSQL 15 + PostGIS (geospatial distance queries) · Redis (rate limiting & hot cache)
AI & Optimisation	Hybrid triage ensemble: deterministic local NLP engine (always runs) + Alibaba Cloud Model Studio Qwen-Max / Google Gemini Flash adapter (optional) · Kuhn-Munkres Hungarian global assignment solver (pure Python, O(n3))
Deployments	Vercel (frontend: web-two-mu-edisuy5tdl.vercel.app) · Render (FastAPI backend: resqgrid-api-9wns.onrender.com) · Supabase / Neon (PostgreSQL + PostGIS) · Upstash (Redis) · Docker Compose (local offline stack)

Three-Week Plan (Buildathon: 20 Sep -- 11 Oct 2026)

Week / Gate	Dates	Core Work	Deliverable & Proof
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Week 1 Gate 1: Problem & Proof	20-26 Sep	5 stakeholder interviews (DMC, Navy, 1990, SLRCS, volunteer) · 32-person survey · architecture design · Pydantic domain schemas · full-stack prototype deployed on Vercel + Render	Gate 1 Write-up PDF + Supporting Evidence PDF uploaded to portal · Live app on Vercel · GitHub repo initialised with commits from 20 Sep
Week 2 Gate 2: Build Checkpoint	27 Sep-3 Oct	Hybrid triage ensemble calibration · PostGIS geospatial matcher · Kuhn-Munkres optimiser · dynamic hazard route penalties · Capacitor Android APK build	Public GitHub repo link · hand-drawn architecture diagram (photo) · 2-min unlisted screen recording of triage-to-dispatch flow
Week 3 Gate 3: Final Submission	4-11 Oct	Offline IndexedDB outbox with idempotent sync · stress testing · 1-2 page AI Usage Report · 12-slide pitch deck · 4-min demo video · Business Case · Declarations	4-min demo video (unlisted YouTube) · full README · live link / APK · Business Case (max 10 pages) · pitch deck PDF · AI Usage Report · Declarations form